

Universal Return Groups and Punctured Rotations

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This paper (Paper XXX of the RIME program) studies the intrinsic raw section dynamics induced by an arbitrary labelled binary-kernel rank- $(n - 1)$ defect. It follows Paper XXIX's canonical rank-five frontier classification but does not import that paper's canonical order law, finite audits, or formalization as premises.

Abstract

Problem. For the canonical single-defect circular family, the missing-image section carries a cyclic return symmetry. For a general labelled defect, most rotation-defect return blocks are only partially defined, and quotienting their dynamics as deterministic maps can destroy the collision guard. The structural source of the canonical symmetry is therefore not apparent from the canonical case alone.

Approach. We work on the rank-five spectator-safe configuration space consisting of a marked pair and three spectators. Every defect event sends a guarded configuration into the missing-image section. The two rotations which move the missing image to the two kernel points produce globally enabled permutation returns g_0, g_1 on that section. Their action defines the universal return group G_d . Partial returns are then projected as labelled relations between G_d -orbits rather than as falsely deterministic quotient maps.

Results. Every labelled binary-kernel rank- $(n - 1)$ defect has two globally enabled section returns and an orientation-independent group

$$G_d = \langle g_0, g_1 \rangle \leq \text{Sym}(Q \setminus \{m_d\}).$$

The resulting relation-valued orbit system preserves and reflects exact section reachability. For the relative return $h_d = g_1 g_0^{-1}$, let Δ be the positive cyclic separation of the two kernel points, put $g = \gcd(n, \Delta)$, and put $\ell = n/g$. We prove that h_d is conjugate to rotation by Δ with one point deleted and the resulting gap closed. Consequently,

$$\text{ctype}(h_d) = (\ell - 1, \ell^{g-1}).$$

If $\gcd(n, \Delta) = 1$, the section contains an intrinsic regular cyclic subgroup C_{n-1} and G_d is transitive. In particular, every such defect at prime ambient size lies in this regime. The canonical cyclic symmetry of Paper XXIX is recovered as the case $g_0 = \text{id}$.

For non-coprime spacing, the remaining branch return has no hidden universal restriction on the fiber with kernel and missing image fixed: as the defect varies in that fiber, the conjugate of g_0 ranges over every permutation a of the punctured carrier. The point-orbits of G_d on D are exactly the connected components of the graph recording which punctured-rotation cycles are joined by a . Every partition of those cycles is realizable by some labelled defect in the same fiber. Fixing the collision image cuts this free parameter to a one-point permutation fiber of size $(n - 2)!$ and imposes an exact forced-join, with one singleton forced-isolation exception, on the realizable point-orbit partitions.

Boundary. The paper gives an exact reduction and a general-defect regime theorem; it does not classify every return group, every guarded quotient edge, or every spectator-safe frontier. It does not establish survivor incidence, transfer membership, projectability, recursive return, credit settlement, or a reset bound. The finite audits are bounded consistency controls bound

into the development closure. They are not proofs of the all- n theorems and are not promoted as independent Computational Certificates.

Keywords. synchronizing automata; circular automata; rank defect; return group; missing-image section; relation-valued quotient; punctured rotation; cycle-gluing graph

Notation Table

Symbol	Meaning
Q	labelled state set, $ Q = n \geq 6$
p	fixed full cycle on Q
$d : Q \rightarrow Q$	labelled rank- $(n-1)$ map with one binary kernel
$K_d = \{k_0, k_1\}$	the unique non-singleton kernel fiber
m_d	the unique missing image of d
$D = Q \setminus \{m_d\}$	image of d and carrier of the universal returns
$\text{Conf}_{2,3}(Q)$	marked pair plus disjoint spectator triple
U_x, H_x	occupied support and hole set of a configuration x
Σ_{m_d}	missing-image section, consisting of configurations with m_d a hole
$\mathcal{R}_t = d \circ p^t$	partial rotation-defect return
t_i	exponent satisfying $p^{t_i}(m_d) = k_i$
g_i	globally enabled permutation return $d \circ p^{t_i} _D$
$G_d = \langle g_0, g_1 \rangle$	universal return group
$\pi_d, \mathcal{N}_d$	orbit map and orbit space Σ_{m_d}/G_d
Δ	positive cyclic separation with $p^\Delta(k_0) = k_1$
$h_d = g_1 g_0^{-1}$	relative return
$H_d = \langle h_d \rangle$	orientation-independent relative cyclic subgroup
$E_0 = Q \setminus \{k_0\}$	branch carrier used to conjugate the return group
ψ_Δ	punctured rotation conjugate to h_d on E_0
a_d	free branch-completion parameter conjugate to g_0
$\Gamma_\Delta(a)$	cycle-gluing graph on the ψ_Δ -cycles
$\mathcal{D}(K, m, c)$	fixed-kernel, fixed-missing-image, fixed-collision defect fiber

1 Introduction

A rank- $(n-1)$ transformation has one local source of noninjectivity: two kernel points share one image. Its dynamics inside a circular automaton are nevertheless not local in the same sense. Away from the kernel, the defect may transport occupied coordinates by an arbitrary labelled bijection. This global transport affects which rank-preserving paths exist before the first strict rank drop.

Paper XXIX solved the complete raw rank-five first-exit problem for the canonical circular family

$$\begin{aligned} Q_n &= \mathbb{Z}/n\mathbb{Z}, & p_n(q) &= q + 1 \pmod{n}, \\ d_n(0) &= d_n(1) = 0, & d_n(q) &= q - 1 \quad (2 \leq q \leq n-1), \end{aligned} \tag{1.1}$$

using a missing-image section, a cyclic return, and a constructive hole-slide normal form [4]. Its section symmetry could appear to be special machinery tied to the order-preserving formula (1.1). The present paper asks for the structure that survives when the labelled defect is arbitrary.

The first answer is universal. Let $K_d = \{k_0, k_1\}$ and let m_d be the missing image. Rotating m_d into either kernel point guarantees that the other kernel point is not occupied. The subsequent defect step is therefore rank-preserving for every state in the missing-image section. Each of these two return blocks acts by a permutation of

$$D = Q \setminus \{m_d\}.$$

They generate the universal return group

$$G_d = \langle g_0, g_1 \rangle \leq \text{Sym}(D).$$

The word *universal* refers to global enabledness on the section. It does not mean that all defects produce isomorphic groups.

The second answer concerns quotienting. The remaining returns $\mathcal{R}_t = d \circ p^t$ are partial: their guards can vary among representatives of a single G_d -orbit. They need not descend to functions on the orbit space. They do, however, descend to existential labelled relations. Since every G_d -orbit is internally controllable by globally enabled positive return words, this relation-valued quotient preserves and reflects reachability exactly.

The third answer identifies a defect-independent part of the return group. For an orientation (k_0, k_1) of the kernel, define

$$h_d = g_1 g_0^{-1}.$$

Although the labelled permutation h_d depends on the inverse branch of d , its conjugacy class does not. It is a punctured rotation whose cycle type is determined only by n and the cyclic kernel separation Δ . This yields the paper's main structural regime:

$$\gcd(n, \Delta) = 1 \implies C_{n-1} \cong H_d \leq G_d, \quad (1.2)$$

where $H_d = \langle h_d \rangle$ acts regularly on D . Thus

canonical cyclic symmetry is not exceptional machinery;
it is the coprime-spacing shadow of a universal relative return.

(1.3)

Here *shadow* is descriptive shorthand for the canonical labelled realization of the regular relative-return cycle. It does not name an additional quotient, representation, or theorem premise.

The theorem spine has three parts.

1. Section 3 constructs the universal returns and proves strong connectivity inside each G_d -orbit.
2. Section 4 proves exact reachability reduction to a relation-valued orbit system and records its spectator-safe first-exit form.
3. Section 5 proves the punctured-rotation normal form and the coprime-spacing regime. Section 6 proves that the other branch return is a free permutation parameter, gives the exact cycle-gluing graph, and classifies its fixed-collision point-orbit partitions. Section 7 recovers (1.1) as a specialization.

The paper remains entirely at the intrinsic raw level. It does not infer a typed transfer, authorization, handoff, or projection witness from a raw path. Those interfaces motivated the all- n question in Paper XXVIII, but they are neither definitions nor premises here [5].

1.1 Related Work and Novelty Boundary

The Cerny conjecture asks for a quadratic reset bound for synchronizing automata [3, 11, 12]. Circular automata admit stronger reset results under hypotheses different from those used here [7]. Almost-group automata, formed from permutation letters and a letter with one nontrivial kernel class, provide a nearby general setting [1]. This paper proves no reset bound and does not classify general almost-group automata.

Recent work on completely reachable almost-group automata studies the nearby class with exactly one defect-one letter and permutation letters, with particular attention to transitive

imprimitive permutation groups [2]. The present paper instead fixes a circular full-cycle carrier and studies its globally enabled return group, the exact punctured-rotation cycle type, and branchwise cycle gluing. It does not give a complete-reachability characterization.

Monotonic and orientable automata isolate order-preserving regimes [8]. Complete and subset reachability instead ask which unlabelled subsets occur as images [6, 9, 10]. The objects here are different in two respects: a legal internal defect step must remain injective on a marked pair together with three spectators, and the quotient retains the marked colors while forgetting only motion internal to a declared universal-return orbit.

Paper XXIX proves a path-free all- n frontier classification for the canonical circular family (1.1) [4]. The new contribution here is not another canonical classification. It is the general-defect construction of G_d , the exact relation-valued reduction supported by its internal controllability, and the punctured-rotation theorem for h_d . The latter explains why the canonical C_{n-1} appears and identifies a larger coprime-spacing regime in which the same regular cyclic subgroup is forced, even when the original p -cyclic order is not preserved by the defect.

2 Raw Section Dynamics

Assume throughout that $n \geq 6$. Let Q be an n -element labelled set, let p be a full cycle on Q , and let

$$d : Q \longrightarrow Q$$

have rank $n - 1$. Such a map has one binary non-singleton fiber

$$K_d = \{k_0, k_1\}, \quad d(k_0) = d(k_1), \quad (2.1)$$

and one missing image m_d . Every other fiber is a singleton. Write

$$D := Q \setminus \{m_d\} = \text{im}(d). \quad (2.2)$$

The temporary orientation (k_0, k_1) will label two returns. Swapping the kernel points exchanges those returns. The groups and orientation-free conclusions below do not change.

2.1 Marked pair, spectators, and holes

Define

$$\text{Conf}_{2,3}(Q) = \{(A, R) : A, R \subseteq Q, |A| = 2, |R| = 3, A \cap R = \emptyset\}. \quad (2.3)$$

The two points of A are the candidate pair for a later strict fusion; the three points of R are spectators. For $x = (A, R)$ put

$$U_x := A \cup R, \quad H_x := Q \setminus U_x. \quad (2.4)$$

The cycle acts without a guard:

$$p(A, R) = (pA, pR). \quad (2.5)$$

The defect acts only when it is injective on the full five-point support:

$$d(A, R) = (dA, dR). \quad (2.6)$$

Because (2.1) is the unique non-singleton fiber, the guard is exactly

$$K_d \not\subseteq U_x \iff H_x \cap K_d \neq \emptyset. \quad (2.7)$$

Thus the spectator triple is part of legality, not an auxiliary decoration.

2.2 Missing-image section and partial returns

The missing-image section is

$$\Sigma_{m_d} = \{x \in \text{Conf}_{2,3}(Q) : m_d \in H_x\}. \quad (2.8)$$

Equivalently, both A and R are subsets of D . Every guarded defect output belongs to this section because no occupied image can equal m_d .

For $t \in \mathbb{Z}/n\mathbb{Z}$, define the partial section return

$$\mathcal{R}_t = d \circ p^t. \quad (2.9)$$

Words act on states from left to right, whereas maps compose from right to left. Thus the written return block $p^t d$ first applies p^t and then d , and its map is exactly $d \circ p^t$ as in (2.9).

Its domain is

$$\text{Dom}(\mathcal{R}_t) = \{x \in \Sigma_{m_d} : p^t H_x \cap K_d \neq \emptyset\}. \quad (2.10)$$

Whenever it is defined, $\mathcal{R}_t(x)$ again belongs to Σ_{m_d} .

3 Universal Returns and the Group G_d

For $i \in \{0, 1\}$, let $t_i \in \mathbb{Z}/n\mathbb{Z}$ be the unique exponent such that

$$p^{t_i}(m_d) = k_i. \quad (3.1)$$

Define

$$g_i := d \circ p^{t_i}|_D. \quad (3.2)$$

Theorem 3.1 (universal return-group theorem).

For every labelled binary-kernel rank- $(n-1)$ defect d :

1. $\mathcal{R}_{t_i}$ is defined on every state of Σ_{m_d} ;
2. $g_i \in \text{Sym}(D)$;
3. on the section,

$$\mathcal{R}_{t_i}(A, R) = (g_i A, g_i R); \quad (3.3)$$

4. the subgroup

$$G_d := \langle g_0, g_1 \rangle \leq \text{Sym}(D) \quad (3.4)$$

is independent of the orientation of K_d ; and

5. *each G_d -orbit in Σ_{m_d} is strongly connected by positive words in the two globally enabled return blocks.*

Proof. Let $x \in \Sigma_{m_d}$. Since $m_d \in H_x$, equation (3.1) gives $k_i \in p^{t_i} H_x$. The guard (2.7) therefore holds after the rotation, so $\mathcal{R}_{t_i}$ is defined for every section state.

The cycle restricts to a bijection

$$p^{t_i} : D \xrightarrow{\sim} Q \setminus \{k_i\}. \quad (3.5)$$

Since K_d is the only non-singleton fiber of d , deletion of either kernel point leaves a bijection

$$d|_{Q \setminus \{k_i\}} : Q \setminus \{k_i\} \xrightarrow{\sim} D. \quad (3.6)$$

Their composition is the permutation g_i , and applying it simultaneously to A and R gives (3.3).

Swapping k_0, k_1 swaps t_0, t_1 and hence swaps the two generators, so G_d is unchanged. Finally, every g_i has finite order. Its inverse is a positive power of itself. Every word in $g_0^{\pm 1}, g_1^{\pm 1}$ is therefore realized by a positive word in globally enabled return blocks. Any two states in one G_d -orbit are mutually reachable.

The theorem does not say that the abstract group or its concrete action is the same for all defects. It identifies the symmetry that every defect supplies from its own two kernel branches.

4 Exact Relation-Valued Orbit Reduction

Let

$$\pi_d : \Sigma_{m_d} \longrightarrow \mathcal{N}_d := \Sigma_{m_d}/G_d \quad (4.1)$$

be the orbit projection. A partial return $\mathcal{R}_t$ need not descend to a function on $\mathcal{N}_d$. Two representatives in one orbit may have different guards, and guarded representatives may enter different output orbits. Requiring a deterministic quotient would add a congruence theorem that is generally unavailable.

Instead define the labelled relation

$$O \xrightarrow{t}_d O' \iff \exists x \in O \cap \text{Dom}(\mathcal{R}_t), \quad \mathcal{R}_t(x) \in O'. \quad (4.2)$$

For $X \subseteq \Sigma_{m_d}$, write

$$x \rightsquigarrow_\Sigma X$$

when a finite sequence of defined section returns sends x into X . Write

$$O \rightsquigarrow_{\mathcal{N}} \pi_d(X)$$

for reachability through (4.2), where the terminal orbit need only intersect X .

Theorem 4.1 (exact relation-valued orbit reduction).

For every $x \in \Sigma_{m_d}$ and every $X \subseteq \Sigma_{m_d}$,

$$x \rightsquigarrow_\Sigma X \iff \pi_d(x) \rightsquigarrow_{\mathcal{N}} \pi_d(X). \quad (4.3)$$

Proof. Projecting a section path gives a path in (4.2).

Conversely, take a quotient path. For its first edge, choose a witness x' in (4.2). The current section state and x' belong to the same G_d -orbit. Theorem 3.1 supplies a positive word in globally enabled returns carrying the current state to x' . Apply the witnessed partial return. Repeat this procedure edge by edge. In the terminal orbit, use universal returns once more to reach the chosen representative in X . The resulting section path lifts the whole quotient path.

The theorem proves preservation and reflection of reachability. It does not assert that π_d is a transition congruence, that an edge in (4.2) has a unique witness, or that $\mathcal{R}_t$ has a single-valued quotient action.

4.1 Spectator-safe target reduction

For a section state $x = (A, R)$, define the section exit set

$$\mathcal{T}_d^\Sigma = \{(A, R) \in \Sigma_{m_d} : \exists u \in \mathbb{Z}/n\mathbb{Z}, \quad p^u A = K_d\}. \quad (4.4)$$

Membership means that a final pure rotation can place the marked pair on the kernel, after which the next defect step is strict. Let $\text{SafeHit}_d(x)$ mean that a word preserves injectivity on the whole marked pair plus spectator triple at every proper defect step and then performs one strict terminal defect step whose fused lineages are the marked pair.

Corollary 4.2 (section Safe-Hit reduction).

For $x \in \Sigma_{m_d}$,

$$\boxed{\text{SafeHit}_d(x) \iff \pi_d(x) \rightsquigarrow_{\mathcal{N}} \pi_d(\mathcal{T}_d^{\Sigma}).} \quad (4.5)$$

Proof. Group every plateau word between consecutive defect events into a return block $p^t d$, represented on the section by $\mathcal{R}_t = d \circ p^t$. The final pure rotation is exactly condition (4.4). Apply Theorem 4.1.

For completeness, the reduction extends to a starting configuration not yet in the section. Define

$$\text{First}_t(x) := \mathcal{R}_t(x) \quad (4.6)$$

whenever the rotation followed by d is a plateau step, equivalently

$$K_d \not\subseteq p^t U_x \iff p^t H_x \cap K_d \neq \emptyset. \quad (4.7)$$

Every such output belongs to Σ_{m_d} .

Corollary 4.3 (first-entry decomposition).

For every $x = (A, R) \in \text{Conf}_{2,3}(Q)$,

$$\begin{aligned} \text{SafeHit}_d(x) \iff & [\exists u, p^u A = K_d] \\ & \vee [\exists t, K_d \not\subseteq p^t U_x \text{ and } \pi_d(\text{First}_t(x)) \rightsquigarrow_{\mathcal{N}} \pi_d(\mathcal{T}_d^{\Sigma})]. \end{aligned} \quad (4.8)$$

Proof. If a safe word contains no plateau defect event before its terminal strict step, its preceding letters form a power of p , giving the first branch. Otherwise isolate its first plateau defect event. Its preceding prefix is p^t ; condition (4.7) holds, the output enters the section, and Corollary 4.2 applies to the remaining return blocks. Conversely, either branch concatenates to a spectator-safe word with the required terminal fusion.

Corollaries 4.2 and 4.3 reduce the raw problem. They do not yet characterize the orbit relation (4.2) for an arbitrary defect.

5 Relative Return and Punctured Rotation

Retain the temporary orientation (k_0, k_1) . Define

$$E_i := Q \setminus \{k_i\}, \quad b_i := d|_{E_i} : E_i \xrightarrow{\sim} D. \quad (5.1)$$

Let $\Delta \in \{1, \dots, n-1\}$ be the positive cyclic separation defined by

$$p^{\Delta}(k_0) = k_1. \quad (5.2)$$

Equivalently, $\Delta = t_1 - t_0 \pmod{n}$. Define the relative return

$$\boxed{h_d := g_1 g_0^{-1} \in G_d.} \quad (5.3)$$

Theorem 5.1 (punctured-rotation normal form).

Put

$$g := \gcd(n, \Delta), \quad \ell := \frac{n}{g}. \quad (5.4)$$

Then:

1. the branch-change bijection

$$\eta : E_1 \longrightarrow E_0, \quad \eta(k_0) = k_1, \quad \eta(q) = q \quad (q \notin K_d) \quad (5.5)$$

satisfies

$$\boxed{b_0^{-1} h_d b_0 = \eta \circ p^\Delta|_{E_0};} \quad (5.6)$$

2. in cyclic coordinates with $k_0 = 0$ and $k_1 = \Delta$, this conjugate is the punctured rotation

$$\psi_\Delta(x) = \begin{cases} x + \Delta, & x + \Delta \not\equiv 0 \pmod{n}, \\ \Delta, & x \equiv -\Delta \pmod{n}, \end{cases} \quad x \in Q \setminus \{0\}; \quad (5.7)$$

3. the cycle type of h_d is

$$\boxed{\text{ctype}(h_d) = (\ell - 1, \ell^{g-1});} \quad (5.8)$$

4. the subgroup

$$\boxed{H_d := \langle h_d \rangle \leq G_d} \quad (5.9)$$

is independent of the orientation of the kernel; and

5. if $\gcd(n, \Delta) = 1$, then h_d is an $(n - 1)$ -cycle on D , so

$$H_d \cong C_{n-1}, \quad G_d = \langle g_0, h_d \rangle \text{ is transitive on } D. \quad (5.10)$$

Proof. From (3.2) and (5.1),

$$g_i = b_i \circ p^{t_i}|_D. \quad (5.11)$$

The inverse of g_0 is

$$g_0^{-1} = p^{-t_0} \circ b_0^{-1},$$

with domains and codomains fixed by (3.5)–(3.6). Hence

$$h_d = b_1 \circ p^{t_1 - t_0} \circ b_0^{-1} = b_1 \circ p^\Delta \circ b_0^{-1}. \quad (5.12)$$

The maps b_0^{-1} and b_1^{-1} are the two inverse branches of d . They agree away from the collision image. More explicitly, $b_0^{-1} b_1 : E_1 \rightarrow E_0$ fixes every point of $Q \setminus K_d$ and maps k_0 to k_1 . Thus

$$b_0^{-1} b_1 = \eta. \quad (5.13)$$

Conjugating (5.12) by b_0 proves (5.6).

Set $k_0 = 0$ and $k_1 = \Delta$. Translation by Δ maps $E_0 = Q \setminus \{0\}$ to $E_1 = Q \setminus \{\Delta\}$. The map η fixes its output unless that output is 0, in which case it maps 0 to Δ . This is (5.7).

Translation by Δ on Q has g cycles, each of length ℓ . The cycle containing 0 is

$$0, \Delta, 2\Delta, \dots, -\Delta.$$

Deleting 0 and replacing the arrow $-\Delta \mapsto 0$ by $-\Delta \mapsto \Delta$ produces one cycle of length $\ell - 1$. The other $g - 1$ cycles remain cycles of length ℓ . This proves (5.8), where the exponent records multiplicity and contributes no ℓ -cycle when $g = 1$.

Swapping k_0, k_1 swaps g_0, g_1 and replaces h_d by

$$g_0 g_1^{-1} = h_d^{-1}.$$

It leaves H_d unchanged. Finally, when $g = 1$, equation (5.8) consists of one cycle of length $n - 1$. Thus H_d acts regularly on D , and every overgroup containing it is transitive. Since $g_1 = h_d g_0$, one also has $G_d = \langle g_0, h_d \rangle$.

Theorem 5.1 removes the off-kernel transport from the conjugacy class of the relative return, not from the entire defect dynamics. The actual labelled permutation h_d still depends on the branch labelling b_0 , and the remaining return g_0 can interact differently with the cycles in (5.8).

Corollary 5.2 (coprime-spacing regime).

If $\gcd(n, \Delta) = 1$, the missing-image section has an intrinsic regular cyclic symmetry $H_d \cong C_{n-1}$. In particular, if n is prime, every labelled binary-kernel rank- $(n - 1)$ defect lies in this regime.

Moreover, Theorem 4.1 remains valid with H_d in place of G_d : each H_d -orbit is strongly connected by positive words in globally enabled return blocks, and the relation-valued quotient

$$\Sigma_{m_d}/H_d \tag{5.14}$$

preserves and reflects section reachability.

Proof. The first statement follows from Theorem 5.1. If n is prime, every $1 \leq \Delta \leq n - 1$ is coprime to n . For the quotient statement, $h_d = g_1 g_0^{-1}$ and h_d^{-1} are positive words in g_0, g_1 because those generators have finite order. Hence every H_d -orbit is internally controllable by globally enabled returns, which is the only group-specific input in the lifting direction of Theorem 4.1.

The quotient (5.14) is a **relative-return necklace quotient**. Its cyclic order is the orbit order of h_d , not necessarily the original p -order. Corollary 5.2 does not classify the remaining guarded return edges or the Safe-Hit target in that cyclic order.

6 Free Branch Completion and Exact Cycle Gluing

Theorem 5.1 fixes the conjugacy class of h_d but leaves open how the other generator meets its cycles. That remaining datum is genuinely free.

Fix p , an oriented kernel $K = \{k_0, k_1\}$, and a missing image m . Let

$$E_0 := Q \setminus \{k_0\}, \quad D := Q \setminus \{m\}, \tag{6.1}$$

and let t_0 satisfy $p^{t_0}(m) = k_0$. Write $\mathcal{D}(K, m)$ for the set of all rank- $(n - 1)$ maps whose unique binary kernel is K and whose missing image is m . For $d \in \mathcal{D}(K, m)$, put

$$b_0 := d|_{E_0} : E_0 \xrightarrow{\sim} D, \quad \boxed{a_d := p^{t_0} \circ b_0 \in \text{Sym}(E_0)}. \tag{6.2}$$

Let ψ_Δ denote the punctured rotation (5.7) on E_0 , expressed in coordinates with $k_0 = 0$ and $k_1 = \Delta$.

Theorem 6.1 (free branch-completion theorem).

The map

$$\Phi_{K, m} : \mathcal{D}(K, m) \longrightarrow \text{Sym}(E_0), \quad d \longmapsto a_d, \tag{6.3}$$

is a bijection. Under conjugation by the branch map b_0 ,

$$\boxed{b_0^{-1}g_0b_0 = a_d, \quad b_0^{-1}h_db_0 = \psi_\Delta, \quad b_0^{-1}G_db_0 = \langle a_d, \psi_\Delta \rangle.} \quad (6.4)$$

Proof. For a given defect, $b_0 : E_0 \rightarrow D$ is a bijection and $p^{t_0} : D \rightarrow E_0$ is a bijection, so (6.2) defines a permutation.

Conversely, let $a \in \text{Sym}(E_0)$ and set

$$b_a := p^{-t_0} \circ a : E_0 \xrightarrow{\sim} D. \quad (6.5)$$

Define $d_a : Q \rightarrow Q$ by

$$d_a(q) = b_a(q) \quad (q \in E_0), \quad d_a(k_0) = b_a(k_1). \quad (6.6)$$

The map b_a is injective with image D . Equation (6.6) introduces exactly one repeated fiber, namely $\{k_0, k_1\}$, and introduces no further image. Thus $d_a \in \mathcal{D}(K, m)$. Its branch map is b_a , and

$$p^{t_0}b_a = a.$$

This construction is inverse to (6.3), because a defect in $\mathcal{D}(K, m)$ is determined by its restriction to E_0 : its omitted value at k_0 must equal the value at k_1 .

Finally, $g_0 = b_0 p^{t_0}|_D$, so

$$b_0^{-1}g_0b_0 = p^{t_0}b_0 = a_d.$$

The middle identity in (6.4) is Theorem 5.1, and the group identity follows from $G_d = \langle g_0, h_d \rangle$.

If the collision image $c = d(k_0) = d(k_1)$ is also fixed, the free parameter is restricted by the single point condition

$$a_d(k_1) = p^{t_0}(c). \quad (6.7)$$

The exact fiber and its gluing consequences are classified after the unrestricted cycle-gluing theorem.

6.1 The cycle-gluing graph

Let $\mathcal{C}_\Delta$ be the set of cycles of ψ_Δ . By Theorem 5.1 it consists of one cycle of length $\ell - 1$ and $g - 1$ cycles of length ℓ . For $a \in \text{Sym}(E_0)$, define the undirected **cycle-gluing graph** $\Gamma_\Delta(a)$ by

$$V(\Gamma_\Delta(a)) = \mathcal{C}_\Delta, \quad (6.8)$$

with an edge between $C, C' \in \mathcal{C}_\Delta$ whenever

$$\exists x \in C, \quad a(x) \in C'. \quad (6.9)$$

Loops may be discarded. The graph records incidence, not a deterministic map on the cycle set: one permutation can send different points of one ψ_Δ -cycle into several cycles.

Theorem 6.2 (exact cycle-gluing theorem).

The orbits of $\langle a, \psi_\Delta \rangle$ on E_0 are exactly the unions

$$\bigcup_{C \in \mathcal{K}} C, \quad (6.10)$$

where $\mathcal{K}$ ranges over the connected components of $\Gamma_\Delta(a)$. Consequently,

$$\boxed{G_d \text{ is transitive on } D \iff \Gamma_\Delta(a_d) \text{ is connected.}} \quad (6.11)$$

Proof. A power of ψ_Δ stays inside one vertex cycle of $\Gamma_\Delta(a)$. An application of a or a^{-1} crosses one of its undirected edges. Hence every group orbit is contained in the union attached to one connected component.

Conversely, suppose C, C' are adjacent. If some $x \in C$ satisfies $a(x) \in C'$, move inside C to x and apply a . Otherwise the edge has a witness $y \in C'$ with $a(y) \in C$; move inside C to $a(y)$ and apply a^{-1} . In either case one can pass from any point of C into C' and then reach every point of C' by powers of ψ_Δ . Repeating along a graph path connects every pair of points in (6.10). This proves the orbit statement. Conjugation by b_0 and (6.4) give (6.11).

The graph therefore answers exactly how g_0 glues the $(\ell - 1, \ell, \dots, \ell)$ cycles. There is no stronger universal gluing law on the unrestricted fiber $\mathcal{D}(K, m)$.

Corollary 6.3 (realization of every cycle partition).

Fix p , the oriented kernel K , and the missing image m . For every set partition $\mathcal{P}$ of $\mathcal{C}_\Delta$, there exists $d \in \mathcal{D}(K, m)$ whose point-orbit partition on D corresponds under b_0 to

$$\left\{ \bigcup_{C \in \mathcal{K}} C : \mathcal{K} \in \mathcal{P} \right\}. \quad (6.12)$$

Proof. For each block $\mathcal{K} \in \mathcal{P}$, choose a cyclic ordering of all points in $\bigcup_{C \in \mathcal{K}} C$ and let $a_{\mathcal{K}}$ be the corresponding full cycle on that union. Let a be the product of these disjoint cycles. Its gluing graph has precisely the blocks of $\mathcal{P}$ as connected components. Theorem 6.1 realizes a by a defect in $\mathcal{D}(K, m)$, and Theorem 6.2 gives (6.12).

When $g > 1$, the discrete partition and the one-block partition in Corollary 6.3 give defects with the same oriented kernel and missing image for which G_d is respectively nontransitive and transitive. Thus the kernel spacing fixes the relative-return cycle type but does not fix how those cycles are joined.

The first constrained fiber is also exact. Fix a collision image $c \in D$ and define

$$\mathcal{D}(K, m, c) := \{d \in \mathcal{D}(K, m) : d(k_0) = d(k_1) = c\}, \quad y_c := p^{t_0}(c) \in E_0. \quad (6.13)$$

Put $s := k_1$, and let $C_s, C_c \in \mathcal{C}_\Delta$ be the ψ_Δ -cycles containing s and y_c , respectively.

Corollary 6.4 (fixed-collision branch fiber and exact gluing constraint).

Restriction of the branch-completion bijection gives

$$\boxed{\mathcal{D}(K, m, c) \xrightarrow{\sim} \{a \in \text{Sym}(E_0) : a(s) = y_c\}, \quad d \mapsto a_d.} \quad (6.14)$$

In particular,

$$|\mathcal{D}(K, m, c)| = (n - 2)!. \quad (6.15)$$

A set partition $\mathcal{P}$ of $\mathcal{C}_\Delta$ is the point-orbit partition of some $d \in \mathcal{D}(K, m, c)$ if and only if

$$\boxed{\begin{aligned} C_s \neq C_c &\implies C_s \text{ and } C_c \text{ lie in the same block of } \mathcal{P}, \\ y_c = s \text{ and } C_s = \{s\} &\implies \{C_s\} \text{ is a block of } \mathcal{P}. \end{aligned}} \quad (6.16)$$

Proof. Equation (6.7) shows that $\Phi_{K, m}$ maps $\mathcal{D}(K, m, c)$ into the right-hand side of (6.14). Conversely, if $a(s) = y_c$, then the inverse construction (6.5)–(6.6) gives

$$d_a(k_0) = d_a(k_1) = p^{-t_0}a(s) = c.$$

This proves (6.14). Since $|E_0| = n - 1$, fixing the image of one point leaves $(n - 2)!$ permutations, proving (6.15).

For necessity in (6.16), if $C_s \neq C_c$, the fixed arrow $s \mapsto y_c$ is an edge joining those two cycle vertices. If $y_c = s$ and $C_s = \{s\}$, then $a(s) = s$ and bijectivity prevents every other point from mapping to s ; hence C_s is isolated in the gluing graph.

For sufficiency, first treat every block of $\mathcal{P}$ not containing C_s as in Corollary 6.3, using one full permutation cycle on the union of its ψ_Δ -cycles. Let $\mathcal{K}_s$ be the block containing C_s . If $y_c \neq s$, the first condition places C_c in $\mathcal{K}_s$ whenever it is distinct from C_s ; choose one full cycle on $\bigcup_{C \in \mathcal{K}_s} C$ in which s is immediately followed by y_c . If $y_c = s$ and $C_s \neq \{s\}$, fix s and choose one full cycle on

$$\left(\bigcup_{C \in \mathcal{K}_s} C \right) \setminus \{s\}.$$

This remaining cycle still meets C_s and every other vertex cycle in $\mathcal{K}_s$, so its gluing graph is connected there. Finally, if $y_c = s$ and $C_s = \{s\}$, the second condition gives $\mathcal{K}_s = \{C_s\}$; fix s . The product of these disjointly supported permutations satisfies $a(s) = y_c$ and has connected components exactly $\mathcal{P}$. Equations (6.14) and Theorem 6.2 realize it by the required defect and identify its point-orbit partition.

7 Canonical Specialization

For the canonical labelled family (Q_n, p_n, d_n) of (1.1),

$$K_{d_n} = \{0, 1\}, \quad m_{d_n} = n - 1. \quad (7.1)$$

Orient the kernel as $(0, 1)$. Then

$$t_0 = 1, \quad t_1 = 2. \quad (7.2)$$

On $D = \{0, \dots, n - 2\}$, direct substitution gives

$$g_0 = \text{id}, \quad g_1(r) = r + 1 \pmod{n - 1}. \quad (7.3)$$

Proposition 7.1 (canonical return group).

For the circular automaton (Q_n, p_n, d_n) ,

$$\boxed{G_{d_n} = H_{d_n} \cong C_{n-1}}. \quad (7.4)$$

The orbit order is the ordinary cyclic order on $D = \mathbb{Z}/(n - 1)\mathbb{Z}$, and the quotient of Theorem 4.1 is the necklace reduction used in Paper XXIX.

Proof. Equation (7.3) shows that $G_{d_n} = \langle g_1 \rangle$. Here $\Delta = 1$, so $h_{d_n} = g_1$ and Theorem 5.1 gives the same regular cyclic subgroup.

The proposition is a specialization, not a premise for the general theory. Conversely, Theorem 5.1 explains the canonical symmetry conceptually: its regular cycle is the simplest labelled realization of the relative return forced throughout the coprime-spacing regime. Paper XXIX adds the further order-preservation and hole-slide arguments needed for a complete canonical frontier classification; those properties do not follow from (7.4) alone.

8 Computational Artifacts

The paper-owned package is available under [experiments/paper30/](#) in the RIME repository.

Surface	Paper-owned path	Role
Bounded consistency control	<code>return_group_audit.py</code> and <code>results/</code>	Exhaustively replays the branch, relative-return, gluing, and fixed-collision definitions for $6 \leq n \leq 8$
Partial formalization	<code>lean/</code>	Checks the data-independent quotient, cycle-gluing, and fixed-fiber implication spine
Development validation	<code>development-manifest.json</code> and <code>validation/</code>	Binds and replays the draft paper-owned closure
Release validation	<code>release-manifest.json</code> , <code>release-environment.json</code> , and <code>validation/</code>	Binds the reader PDF and nested development closure; records local closure verification

The retained finite audits are bounded consistency controls bound into the development closure. They cover 40,200 branch-completion cases and 110 fixed-collision fibers. They are not proofs of the all- n theorems and are not promoted as independent Computational Certificates. The Lean development does not formalize the punctured-rotation arithmetic or the constructive permutation realizations. The nested development manifest claims no publication identity. The outer release receipt records local closure verification, not independent mathematical validation.

9 Claim Status and Boundary

Surface	Status	Scope
Globally enabled returns g_0, g_1	Theorem 3.1	Every labelled binary-kernel rank- $(n - 1)$ defect, $n \geq 6$
Orientation-independent return group G_d	Theorem 3.1	Same scope
Strong connectivity inside each G_d -orbit	Theorem 3.1	Positive words in the universal return blocks
Relation-valued orbit reachability reduction	Theorem 4.1	Every subset of the missing-image section
Section Safe-Hit reduction	Corollary 4.2	Raw marked-pair plus three-spectator dynamics
First-entry decomposition	Corollary 4.3	Arbitrary raw 2 + 3 starting state
Punctured-rotation normal form	Theorem 5.1	Every labelled defect in the declared scope
Coprime-spacing regular cyclic subgroup	Corollary 5.2	$\gcd(n, \Delta) = 1$; all defects when n is prime
Free branch-completion parameter	Theorem 6.1	Fixed oriented kernel and missing image; collision image not fixed
Exact cycle-gluing graph	Theorem 6.2	Every labelled defect in the declared scope
Realization of every relative-cycle partition	Corollary 6.3	Fixed oriented kernel and missing image; collision image may vary
Fixed-collision branch fiber and cardinality	Corollary 6.4	Fixed oriented kernel, missing image, and collision image
Exact fixed-collision point-orbit partitions	Corollary 6.4	Forced join, with the stated singleton forced-isolation exception
Canonical cyclic specialization	Proposition 7.1	The circular family (Q_n, p_n, d_n) of (1.1)
Classification of abstract groups G_d	Open	Not determined by the point-orbit partition alone
General orbit-edge or Safe-Hit characterization	Open	Remaining guarded returns on the concrete orbit quotient
Complete survivor-placement frontier	Open	Requires terminal incidence after pair hitting
Raw-to-typed transfer/projectability bridge	Open	No new semantic fields are introduced here

The theorem proofs are data-independent. The finite audits are bounded consistency controls bound into the development closure. They are not proofs of the all- n theorems, are not manuscript premises, and are not promoted as independent Computational Certificates.

The following implications are explicitly excluded:

$$G_d \cong G_{d'} \implies \text{equal spectator-safe frontiers,} \quad (8.1)$$

$$G_d \text{ transitive on } D \implies \text{automatic Safe-Hit,} \quad (8.2)$$

and

$$\text{raw Safe-Hit} \implies \text{declared transfer membership or projectability.} \quad (8.3)$$

The quotient theorem uses the concrete group action and the concrete relation (4.2), not merely the abstract isomorphism type of G_d .

10 Conclusion

Every labelled binary-kernel rank- $(n - 1)$ defect induces two globally enabled permutation returns on its missing-image section. Their group supplies exact internal controllability, which is

precisely what permits partial return dynamics to be reduced as a relation without pretending that the orbit map is a deterministic congruence.

The relative return between the two kernel branches has a stronger universal form. Up to branch conjugacy it is rotation by the kernel separation with one point deleted and the gap closed. Its cycle type is therefore fixed by (n, Δ) , and coprime kernel spacing forces a regular cyclic subgroup of order $n - 1$. Outside that regime, and on the fiber with kernel and missing image fixed, the other branch return is a free permutation parameter: its incidence graph on the punctured-rotation cycles determines the point-orbits of G_d on D exactly, and every partition of those cycles is realizable. Fixing the collision image cuts the free branch to a one-point permutation fiber and yields the exact forced-join and singleton-isolation rules of Corollary 6.4. The cyclic mechanism isolated in the canonical family is thus a special visible case of a general branch-relative symmetry.

10.1 Outlook

The next general-defect questions begin after, not inside, the present theorem spine. The point-orbit partition on D is now explicit both on the broad fixed- (K, m) fiber and on its fixed-collision subfibers. The remaining group-theoretic question is which abstract subgroups $\langle a, \psi_\Delta \rangle$ and coloring orbits matter under constraints beyond (K, m, c) , such as order-preserving transport, parity, or a prescribed off-kernel table. The dynamical question is whether the remaining existential quotient edges admit path-free normal forms. A full frontier theorem would additionally need the terminal survivor-incidence map. The separate raw-to-typed bridge remains an interface theorem: neither group symmetry nor raw reachability supplies it automatically.

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